

DSL Compiler Simulation – Conditional Expression

1. Problem Understanding

This project simulates a simple Domain Specific Language (DSL) compiler that evaluates a **conditional expression** using the ternary operator.

The DSL program:

max = (a > b) ? a : b

is used to determine the maximum of two numbers. The goal is to demonstrate how a compiler processes this statement through different phases such as lexical analysis, parsing, semantic analysis, and code generation.

2. Input Program

a = 8

b = 5

max = a > b ? a : b

3. Compiler Phases

Lexical Analysis

- Identifies tokens:
 - Identifiers → a, b, max
 - Numbers → 8, 5
 - Operators → >
 - Conditional symbols → ?, :

Syntax Parsing

- Recognizes the structure:

max = (a > b) ? a : b

Semantic Analysis

- Checks:
 - Variables are valid ✓
 - Data types are integers ✓
 - Condition is logically correct ✓

Intermediate Code Generation

if a > b goto L1

t1 = b

goto L2

L1: t1 = a

L2: max = t1

4. Final Output

max = 8

5. Logic Used

- If $a > b$, assign a to max
- Otherwise, assign b to max
- Implemented using an if-else condition in C

6. Justification of Feature

The ternary conditional operator is chosen because:

- It represents real compiler decision-making
- Demonstrates branching logic
- Easy to simulate in compiler phases
- Commonly used in programming languages

Conclusion

This project demonstrates how a compiler evaluates conditional expressions step-by-step. It provides a clear understanding of tokenization, parsing, semantic checking, and execution in a simplified manner.